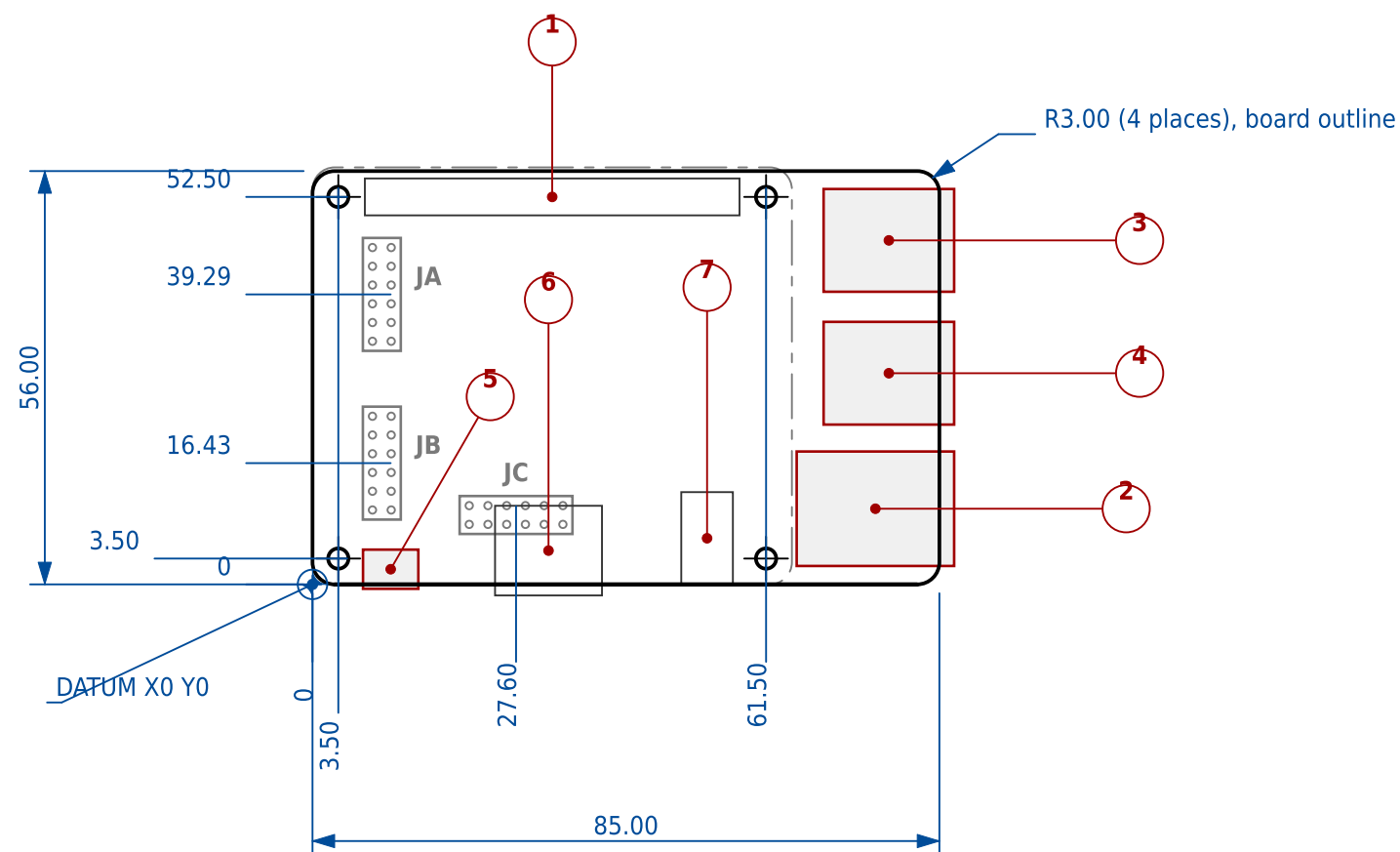




## NOTES

1. All dimensions in millimetres. Datum is the lower-left corner of the board outline; X right, Y up, viewed from the component side.
2. Phantom outline is the Digilent Pmod HAT Adapter fitted to the 40-pin GPIO header, drawn in this board's frame because its mounting holes coincide with this board's.
3. Connector outlines are the component body as drawn by Raspberry Pi Ltd, overhang past the board edge included.
4. Mounting hole diameter: Drawing note, quoted: 4x M2.5 MOUNTING HOLES DRILLED TO 2.75 +/- 0.05mm.
5. Hole IDs are assigned by this drawing, bottom row first then left to right, and mean the same thing on every Raspberry Pi sheet here.

6. No mounting hole keep-out is dimensioned on this model's own drawing. Design to the 6.2 mm the Raspberry Pi HAT mechanical specification ([github.com/raspberrypi/hats](https://github.com/raspberrypi/hats)) requires around each hole.
7. Hole positions are the 3.5 mm inset and 58 x 49 mm rectangle dimensioned on this model's drawing.
8. Pmod host JC on the Pmod HAT Adapter overhangs the lower board edge in the same region as this board's HDMI and power connectors. Digilent's own reference manual warns to fit the two standoffs opposite the 40-pin connector so that JC's pins cannot touch the HDMI shell.
9. Those host positions are DERIVED, not published: Digilent issue no mechanical drawing for the adapter. Good to about  $\pm 0.75$  mm. JA and JB face out of the left edge, JC out of the lower edge, all right-angle hosts whose bodies overhang the edge. The Pmod HAT Adapter sheet has the derivation.

## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm       | KEEPOUT mm |
|-----|-------|-------|--------------|------------|
| MT1 | 3.50  | 3.50  | 2.75 +/-0.05 | not given  |
| MT2 | 61.50 | 3.50  | 2.75 +/-0.05 | not given  |
| MT3 | 3.50  | 52.50 | 2.75 +/-0.05 | not given  |
| MT4 | 61.50 | 52.50 | 2.75 +/-0.05 | not given  |

## FEATURE SCHEDULE

| # | FEATURE                     | X EXTENT mm    | Y EXTENT mm    |
|---|-----------------------------|----------------|----------------|
| 1 | 40-pin GPIO header          | 7.10 to 57.90  | 50.00 to 55.00 |
| 2 | Ethernet RJ45               | 65.65 to 87.00 | 2.50 to 18.00  |
| 3 | USB 2.0 type A (upper pair) | 69.30 to 87.00 | 39.65 to 53.57 |
| 4 | USB 2.0 type A (lower pair) | 69.30 to 87.00 | 21.65 to 35.57 |
| 5 | micro-USB power input       | 6.85 to 14.35  | -0.60 to 4.71  |
| 6 | HDMI type A                 | 24.75 to 39.25 | -1.50 to 10.65 |
| 7 | 3.5 mm A/V jack             | 50.00 to 57.00 | 0.00 to 12.50  |

## PMOD HAT ADAPTER HOSTS

| PORT | EDGE        | CX mm | CY mm | PIN1 X mm | PIN1 Y mm |
|------|-------------|-------|-------|-----------|-----------|
| JA   | left edge   | 9.40  | 39.29 | 8.13      | 45.60     |
| JB   | left edge   | 9.40  | 16.43 | 8.13      | 22.70     |
| JC   | bottom edge | 27.60 | 9.40  | 21.25     | 8.13      |

## SOURCES

1. Mechanical drawing: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.dxf> - Raspberry Pi Ltd
2. Mounting holes: <https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-mechanical-drawing.pdf> - Drawing note, quoted: 4x M2.5 MOUNTING HOLES DRILLED TO 2.75 +/- 0.05mm.

## RASPBERRY PI 3 MODEL B

85 x 56 mm

|                           |                                                                            |                             |                     |            |
|---------------------------|----------------------------------------------------------------------------|-----------------------------|---------------------|------------|
| DRAWN<br><b>generated</b> | DATE<br><b>2026-09-10</b>                                                  | UNITS<br><b>mm</b>          | SCALE<br><b>1:1</b> |            |
| SIZE<br><b>A3</b>         | SHEET<br><b>1 OF 1</b>                                                     | DRAWING NO<br><b>RPI-02</b> | REV<br><b>A</b>     |            |
| MATERIAL<br><b>PCB</b>    | GENERAL TOLERANCE<br><b>edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08</b> |                             |                     | PROJECTION |